* Date of Revision: October 2022
Ca^{++}-antagonist

MAL

HERBESSER® 90 SR

(Diltiazem hydrochloride)
Sustained Release Capsules

Mitsubishi Tanabe Pharma

CONTRAINDICATIONS (HERBESSER is contraindicated in the following patients.)

(1) Patients with severe congestive cardiac failure [Symptoms of cardiac failure may be aggravated].
(2) Patients with second or third degree atrioventricular block or sick sinus syndrome (persistent sinus bradycardia) (less than 50 beats/ minute), sinus arrest, sinoatrial block, etc.) [Depression of cardiac stimulation and cardiac conduction may occur excessively].
(3) Patients with a history of hypersensitivity to any of the ingredients of this product.
(4) Pregnant women and women who may possibly be pregnant [See "Use during Pregnancy, Delivery or Lactation" section].
(5) Patients receiving asunaprevir-containing product, ivabradine hydrochloride, or lomitapide mesylate [See "Drug Interactions" section].

DESCRIPTION

- Diltiazem hydrochloride occurs as white crystals or crystalline powder. It is odorless.
- It is very soluble in formic acid, freely soluble in water, in methanol and in chloroform, sparingly soluble in acetonitrile, slightly soluble in dehydrated ethanol and in acetic anhydride, and practically insoluble in ether.
- Melting point: 210 ºC – 215 ºC (decomposition)

Product's name	Content of Diltiazem hydrochloride	Description of the product
HERBESSER® 90SR	90 mg per capsule	White sustained – release capsule

INDICATIONS

- Hypertension. It may be used alone or in combination with other antihypertensive medications, such as diuretics.
- Angina pectoris, variant angina

DOSAGE AND ADMINISTRATION

For adults, 1 capsule (90mg as diltiazem hydrochloride) twice a day orally. The dosage may be increased or decreased according to the severity of symptoms. Diltiazem hydrochloride has an additive antihypertensive effect when used with other antihypertensive agents. Therefore, the dosage of diltiazem hydrochloride or the concomitant antihypertensive may need to be adjusted when adding one to the other.

PRECAUTIONS

1. Careful Administration (HERBESSER should be administered with care in the following patients.)

1) Patients with congestive cardiac failure. [Symptoms of cardiac failure may be aggravated.]
2) Patients with severe bradycardia (less than 50 beats/min) or first grade atrioventricular block. [Depression of cardiac stimulation and cardiac conduction may occur excessively.]

3) Patients with excessively low blood pressure [Blood pressure may be further decreased.]
4) Patients with severe hepatic or renal impairment. [The metabolism and excretion of this product may be prolonged, and effects may be intensified.]

2. Important Precautions

1) It has been reported that abrupt cessation of calcium antagonists may aggravate the patient's symptoms. In case where cessation of this product is required, the dosage should be reduced gradually under careful observation of the patient. Patients should be instructed not to discontinue taking this product without consulting physicians.
2) Since dizziness, etc. may occur due to antihypertensive action of this product, patients should be cautioned against engaging in potentially hazardous activities, such as working at altitude or driving motor vehicles.
3) Prolonged QT and ventricular arrhythmia have been reported in coadministration of terfenadine with other antiarrhythmic agents (disopyramide phosphate)

3. Drug Interactions

This product is metabolized mainly by cytochrome P450 3A4 (CYP3A4) metabolizing enzyme.

(1) Contraindications for Co-administration (Do not co-administer with the following.)

Drugs	Signs, Symptoms, and Treatment	Mechanism and Risk Factors
Asunaprevir Daclatasvir hydrochloride/ asunaprevir / beclabuvir hydrochloride	The blood concentration of asunaprevir increased. Hepatobiliary disorder may occur and become severe.	This product inhibits CYP3A, and the metabolism of these drugs is inhibited.
Ivabradine hydrochloride	Excessive bradycardia may occur.	This product inhibits CYP3A, the metabolism of ivabradine is inhibited, and the blood concentration of ivabradine is increased. The heart rate reducing effect of ivabradine hydrochloride is potentiated additively.
Lomitapide mesylate	The blood concentration of lomitapide mesylate may be markedly increased.	This product inhibits CYP3A, and the metabolism of lomitapide is inhibited.

(2) Precautions for coadministration (HERBESSER should be administered with care when co-administered with the following drugs.)

Drugs	Signs, Symptoms, and Treatment	Mechanism and Risk Factors
Drugs with antihypertensive effects (antihypertensive drugs, nitric acid preparation, etc.)	Antihypertensive effects may be intensified. Blood pressure should be measured periodically to adjust the dosage.	Antihypertensive effects may be intensified additively.
Beta blockers (bisoprolol fumarate, propranolol hydrochloride, atenolol, etc.)	Bradycardia, atrioventricular block, sinoatrial block, etc. may occur. Pulse rate should be measured periodically, and electrocardiogram should be performed as needed. If any abnormalities are observed, the dosage should be reduced or	Depression of cardiac stimulation and cardiac conduction, negative inotropic effects, and antihypertensive effects may be intensified additively. Particular attention should be given to triple therapy using this product with digitalis
Rauwolfia preparations (reserpine, etc.)		

	administration should be discontinued.	preparation and beta blocker or rauwolfia preparation.
Digitalis preparations (digoxin, methyldigoxin)	Bradycardia, atrioventricular block, etc. may occur. In addition, toxic symptoms (nausea, vomiting, headache, dizziness, abnormal vision, etc.) including above arrhythmic symptoms may occur due to an increased blood concentration of digitalis preparation. Presence or absence of digitalis toxicity should be observed periodically, and electrocardiogram should be performed. In addition, blood concentration of digitalis preparation should be measured as needed. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	Depression of cardiac stimulation and cardiac conduction may be intensified additively. Particular attention should be given to triple therapy using this product with digitalis preparation and beta blocker. This product may be increase blood concentrations of digitalis preparations.
Antiarrhythmic agents (amiodarone hydrochloride, mexiletine hydrochloride, etc.)	Bradycardia, atrioventricular block, sinus arrest, etc. may occur. Pulse rate should be measured periodically, and electrocardiogram should be performed as needed. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	Depression of cardiac stimulation and cardiac conduction may be intensified additively.
Fingolimod hydrochloride	Severe bradycardia or heart block may occur by concomitant use of this product during the initiation of fingolimod by hydrochloride.	Both diltiazem hydrochloride and fingolimod hydrochloride may induce bradycardia or heart block.
Aprindine hydrochloride	Symptoms (bradycardia, atrioventricular block, sinus arrest, tremor, dizziness, light-headedness, etc.) may occur due to increased blood concentrations of both drugs. Clinical symptoms should be observed periodically, and electrocardiogram should be performed as needed. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	Each drug may affect a common metabolizing enzyme (cytochrome P450), and increase blood concentration of each drugs
Dihydropyridine calcium antagonists (nifedipine, amlodipine besilate, etc.)	Symptoms (intensified antihypertensive effects, etc.) may occur due to increased blood concentration of dihydropyridine calcium antagonist. Clinical symptoms should be observed	This product may inhibit the metabolizing enzyme (cytochrome P450) of these drugs, and increase their blood concentrations.

	periodically. If any abnormalities are observed the dosage should be reduced or administration should be discontinued.	
Simvastatin	Rhabdomyolysis or myopathy may occur due to increased blood concentration of simvastatin. Clinical symptoms should be observed periodically. If any abnormalities are observed, administration should be discontinued.	
Triazolam	Symptoms (prolonged sleeping time, etc.) may occur due to increased blood concentration of triazolam. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Midazolam	Symptoms (intensified sedative and hypnotic effect, etc.) may occur due to increased blood concentration of midazolam. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Carbamazepine	Symptoms (sleepiness, nausea, vomiting, dizziness, etc.) may occur due to increased blood concentration of carbamazepine, Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Selegiline hydrochloride	Effects and toxicity of selegiline hydrochloride may be intensified. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Theophylline	Symptoms (nausea, vomiting, headache, insomnia, etc.) may occur due to increased blood concentration of theophylline. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or	

	administration should be discontinued.	
Cilostazol	Effects of cilostazol may be intensified. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Apixaban	Effects of apixaban may be intensified. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Vinorelbine tartrate	Effects of vinorelbine tartrate may be intensified. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Cyclosporin	Symptoms (renal disorder, etc.) may occur due to increased blood concentration of cyclosporin. Clinical symptoms should be observed periodically, and blood concentration of cyclosporin should be measured. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Tacrolimus hydrate	Symptoms (renal disorder, etc.) may occur due to increased blood concentration of tacrolimus. Clinical symptoms should be observed periodically, and blood concentration of tacrolimus should be measured. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	
Phenytoin	Symptoms (ataxia, dizziness, nystagmus, etc.) may occur due to increased blood concentration of phenytoin. Clinical symptoms should be observed periodically. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued. Effect of this product may be attenuated.	This product may inhibit the metabolizing enzyme (cytochrome P450) of phenytoin. and increase blood concentrations of phenytoin. In addition: phenytoin may stimulate metabolism of this product, and decrease blood concentration of this product
Cimetidine	Symptoms (intensified antihypertensive effect, bradycardia, etc.) may occur due	These drugs may inhibit the metabolizing enzyme (cytochrome P450) of this

HIV protease inhibitors (ritonavir, saquinavir mesylate, etc.)	to increased blood concentration of this product. Clinical symptoms should be observed periodically, and electrocardiogram should be performed as needed. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	product, and increase blood concentration of this product.
Rifampicin	Effects of this product may be attenuated. Clinical symptoms should be observed periodically, and if possible, blood concentration of this product should be measured. If any abnormalities are observed, appropriate therapeutic measures such as changing to other drugs or increasing the dosage of this product should be taken.	Rifampicin may induce the metabolizing enzyme (cytochrome P450) of this product, and decrease blood concentrations of this product.
Anesthetics drugs (isoflurane, enflurane, halothane, etc.)	Bradycardia, atrioventricular block, sinus arrest, etc. may occur. Electrocardiogram should be monitored. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	Depression of cardiac stimulation and cardiac conduction may be intensified additively.
Muscle relaxants (pancuronium bromide, vecuronium bromide, etc.)	Effects of muscle relaxants may be intensified. Caution should be exercised to muscle relaxants action. If any abnormalities are observed, the dosage should be reduced or administration should be discontinued.	This product may inhibit the acetylcholine release from the presynaptic terminals at the neuromuscular junction.

4. **Adverse Reactions**

   432 cases (4.5%) of adverse reactions were reported out of total 9,347 cases. The most common occurrences as well as their frequency of presentation are: Gastrointestinal system 1.3% (stomach discomfort 0.2%, constipation 0.2%, abdominal pain 0.1%, etc.), cardiovascular system 1.3% (bradycardia 0.4%, dizziness 0.4%, hot flush of face 0.2%, A-V block 0.2% etc.), hypersensitivity 1.2%, headache 0.2%, etc.

   (1) **Clinically significant adverse reactions (rarely: <0.1%, no adverb: incidence unknown because of spontaneous reports)**

      1) **Complete atrioventricular block, severe bradycardia** (early symptom: bradycardia, dizziness, light-headedness, etc.) may occur rarely. If any abnormalities are observed, administration should be discontinued, and atropine sulfate hydrate, isoprenaline, etc. should be administered or appropriate therapeutic measures such as cardiac pacing should be taken as needed.

      2) **Congestive cardiac failure** may occur. If any abnormalities are observed, administration should be discontinued, and appropriate therapeutic measures such as administration of cardiotonic drug, etc. should be taken.

      3) **Oculomucocutaneous syndrome (Steven-Johnson syndrome), toxic epidermal necrolysis (Lyell syndrome), erythroderma (exfoliative**

dermatitis), acute generalized exanthematous pustulosis may occur. If erythema, blister, pustule, pruritis, fever, enanthema, etc. occur, administration should be discontinued, and appropriate therapeutic measures should be taken.

4) **Hepatic function disorder** or **jaundice** with increased AST (GOT), ALT (GPT) or ɤ-GTP may occur. The patient's conditions should be observed carefully. If any abnormalities are observed, administration should be discontinued, and appropriate therapeutic measures should be taken.

(2) Other adverse reactions

If any adverse reactions are observed, appropriate therapeutic measures such as discontinuation of treatment should be taken.

Incidence / Type	5% > ≥ 0.1%	< 0.1%	Incidence unknown
Cardiovascular	Bradycardia, atrioventricular block, facial flushing, dizziness	Sinus arrest, decreased blood pressure, palpitation, chest pain, oedema	Sinoatrial block
Psychoneurotic	Malaise, headache, headache dull	Cramps in the calves, feelings of weakness, sleepiness, insomnia	Parkinsonian-like symptom
Hepatic	Increased AST (GOT), increased ALT(GPT)	Jaundice	Increased Al-P, increased LDH, increased ɤ-GTP, hepatic hypertrophy
Hypersensitivity	Rash	Pruritis, multiforme erythematous, rash, urticaria	Photosensitivity, pustule
Gastrointestinal	Stomach discomfort, constipation, abdominal pain, heartburn, anorexia, nausea	Soft stool, diarrhea, thirst.	----------
Hematologic	----------	----------	Decreased platelets count, decreased white blood cell count
Other	----------	----------	Gingival hypertrophy, gynaecomastia: numbness

5. Use in the Elderly

Excessive decrease in blood pressure is generally considered undesirable in elderly patients. Therefore, HERBESSER should be administered with care such as starting from a lower dosage while, carefully monitoring the patient's condition.

6. Use during Pregnancy, Delivery or Lactation

1) HERBESSER should not be administered to pregnant women or women who may possibly be pregnant. [Animal studies have shown teratogenicity (mice: skeletal abnormality, appearance abnormality) and embryotoxicity (mice, rats: fatality).]

2) Use of this product in lactating women is not recommended. If treatment with this product is judged to be essential, breast feeding must be discontinued during treatment. [It has been reported that diltiazem is excreted in human breast milk.]

7. **Pediatric Use**
   The safety of HERBESSER in children has not been established.
8. **Overdosage**
   Symptoms:
   Bradycardia, complete atrioventricular block, cardiac failure, hypotension, etc. may occur after overdosage of HERBESSER. These symptoms have been also reported as adverse reactions.
   Treatment:
   In case of overdosage, administration should be discontinued, and gastric lavage should be performed to remove the product as needed, and the following appropriate therapeutic measures should be taken.
   1) Bradycardia, complete atrioventricular block
      Atropine sulfate hydrate, isoprenaline, etc. should be administered or cardiac pacing should be taken.
   2) Cardiac failure, hypotension
      Cardiotonic drug, vasopressor, infusion, etc. should be administered or assisted circulation should be performed.

PHARMACOKINETICS

1. **Plasma level**
   When 1 capsule (90 mg as diltiazem hydrochloride) of HERBESSER 90 SR was orally administered to healthy adult men, its plasma level reached the peak (about 40 ng/ml) about 7 hours after administration.[1)]

   The plasma elimination half life was about 8.4 hours. In case of repeated administration of HERBESSER 90 SR twice a day to healthy adult men, plasma level at 1 to 10 hours after administration was about 91 ng/ml.[2)]

2. **Metabolism**
   When HERBESSER 90 SR was administered to healthy adult men, deacetyl diltiazem, deacetyl-N-monodemethyl diltiazem, deacetyl-0-demetyl diltiazem, deacetyl-N, 0-demetyl diltiazem and N-monodemethyl diltiazem were detected in urine as metabolites. A part of these metabolites are conjugated with glucuronic acid or sulfuric acid in body. [3)]

CLINICAL STUDIES

1. **Hypertension**
   Usefulness of HERBESSER 90 SR in the treatment of essential hypertension was proved by four double-blind comparative studies with placebo, reserpine and propranolol as control drugs.[4 – 7)]
2. **Angina pectoris**
   Usefulness of HERBESSER 90 SR in the treatment of anginal pain due to effort of angina and old myocardial infarction was proved by two double-blind comparative studies including a placebo-controlled study.[8,9)]

PHARMACOLOGY

The therapeutic benefits achieved with HERBESSER 90 SR such as improvement of myocardial ischemia and hypotensive effect are believed to be related to its ability to dilate vessels by inhibiting the influx of calcium ion into coronary and peripheral vessels of smooth muscle cell.

1. **Action on blood pressure**
   (1) Lower the elevated blood pressure gradually although hardly affects the normal blood pressure (rat, human).[10-12]
   Suppresses the elevation of blood pressure induced by exercise load (human).[13]
   (2) Lower blood pressure without decreasing cerebral and renal blood flow (dog, human).[14-17]
   (3) Suppresses myocardial and vascular hypertrophy together with lowering blood pressure (rat).[18]
2. **Effects on myocardial ischemia**
   (1) Increases coronary blood flow into myocardial ischemic region by dilating the collateral channels and large coronary artery (dog).[19-22]
   (2) Suppresses coronary artery spasms (monkey, human).[23,24]
   (3) Decreases myocardial oxygen consumption without decreasing cardiac output by decreasing after-load and heart rate due to peripheral vasodilating effect (dog).[25]
   (4) Retains cardiac function and myocardial energy metabolism and reduces the infarct size by inhibiting extra calcium ion influx during myocardial ischemia (rat).[26]
3. **Action on sinus rhythm and cardiac conduction system**
   Slightly prolong spontaneous sinus rhythm interval and A-V node conduction time. Does not effect on His-Purkinje conduction time (dogs and human).[25, 27, 28]

PRECLINICAL STUDIES

1. **Toxicity**
   (1) Acute toxicity (LD 50 mg/kg)[29]

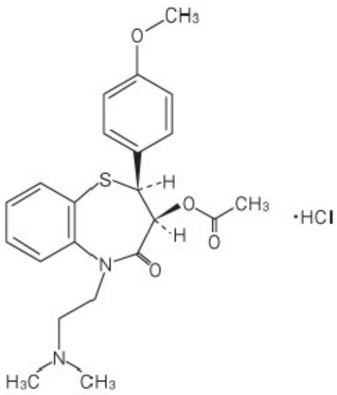
Animal	Route	p.o.		s.c.		i.v.	
	Sex	Men	Women	Men	Women	Men	Women
ddY – strain mouse		740	640	260	280	61	58
Wistar – strain rat		560	610	520	550	38	39

   (2) Chronic toxicity
   When 2, 10 and 25 mg/kg/day each of diltiazem hydrochloride were given orally to SD – strain rated and 5, 10 and 20mg/kg/day to beagle dogs, for successful six months respectively, general state as well as urinary and histopathological findings of these animals were not significantly different from those of the control. Hematological findings in dogs given orally 20 mg/kg of diltiazem hydrochloride revealed a rise of GPT but it was transient and tended to recover the normal level at the end of experiment.[30]

2. **Teratogenicity**
   The effect of diltiazem hydrochloride on the fetus was examined by the method as specified in "Policy for Assurance of Drug Safety" notified by the Ministry of Health and Welfare of Japan. At the oral dose level of more than 10 mg/kg in ICR – JCL – strain mice and more 200 mg/kg in Wistar – strain rats, diltiazem hydrochloride caused death of the fetus. At the oral dose level of more than 50 mg/kg in ICR – JCL strain mice, diltiazem hydrochloride provoked teratogenic effect. At an oral dose of 400 mg/kg in Wistar – strain rats, diltiazem hydrochloride did not provide teratogenic effects.[31]

PHYSICOCHEMISTRY
Nonproprietary name:
Diltiazem Hydrochloride (JAN)
Diltiazem (INN)

Chemical name:
(2S,3S)-5-[2-(Dimethylamino) ethyl]-2-(4-methoxyphenyl)-4-oxo-2,3,4,5-tetrahydro-1,5-benzothiazepin-3-yl-acetate monohydrochloride



$C_{22}H_{26}N_2O_4S \cdot HCl$: 450.98

Description:

- Diltiazem hydrochloride occurs as white crystals or crystalline powder. It is odorless.
- It is very soluble in formic acid, freely soluble in water, in methanol and in chloroform, sparingly soluble in acetonitrile, slightly soluble in acetic anhydride and in ethanol (99.5), and practically insoluble in diethyl ether.
- Melting point: 210 - 215°C (with decomposition)

STORAGE AND HANDLING
Storage : Store in a tight and light-resistant container. Do not store above 30°C.

Caution : Dispense by physician's prescription or direction
Keep out of reach from children
Swallow the capsules without chewing or opening

MODE OF ADMINISTRATION
Oral

PACKAGING
Box of 10 blister sheets of 10 capsules

REFERENCES

1) A. Darragh et al.: (Unpublished)
2) A. C. Houston et al. (Unpublished)
3) J. Sugihara et al.: J. Pharmacobiodyn.,7, 24 (1984)
4) S. Yorifuji et al.: J. Adult Disease,9, 893 (1979)
5) M. Ikeda et al.: lgaku No Ayumi,110, 302 (1979)
6) T. Watanabe et al.: lgaku No Ayumi,120, 854 (1962)
7) M. Ikeda et al.: lgaku No Ayumi,121, 222 (1962)
8) Y Mizuno et al.: Jpn. J. Clin. Exp. Med.,50, 565 (1973)
9) K. Kasahara et al.: Diagnosis & Treatment,63, 696 (1975)
10) M. Sato et al.: Folia pharmacol. japon.,75, 99 (1979)
11) I. Yamaguchi et al.: Folia pharmacol. japon.,75, 191 (1979)
12) K. Aoki et al.: Eur. J. Clin. Pharmacol.,25, 475 (1983)

13) T. Yamakado et al.:Am J. Cardiol.,52, 1023 (1983)
14) S. Murata et al.: Jpn. J. Pharmacol.,32, 1033 (1982)
15) I. Yamaguchi et al.: Jpn. J. Pharmacol.,24, 511 (1974)
16) Y, Kuriyama et al.: J. Jpn. Coll. Angiol.,27, 89 (1987)
17) J. Choki et al.: J. Jpn. Coll. Angiol.,26, 1297 (1986)
18) H. Iwasaki et al.: Jpn. J. Vet. Sci.,46, 323 (1984)
19) M. Sato et al: Arzneimittelforsch.,21, 1338 (1971)
20) K. Takeda et al.: Jpn. Heart J.,18, 92 (1977)
21) T. Nagao et al.: Jpn. J. Pharmacol.,25, 281 (1975)
22) M. Nakamura et al.: Chest.,78, 205 (1980)
23) N. Taira et al.: Circ. Res.,52,1 (1983)
24) H. Yasue et al.: J. Clin. Sci.,21, 597 (1985)
25) T. Nagao et al.: Folia pharmacol. japon.,77, 195 (1981)
26) A. Zamanis et al.: J. Mol. Cell. Cardiol.,14, 53 (1982)
27) H. Nakaya et al.: Folia pharmacol. japon.,76, 697 (1980)
28) C. Kawai et al.: Circulation,63, 1035 (1981)
29) T. Nagao et al.: Jpn. J. Pharmacol.,22, 467 (1972)
30) T. Fujita et al.: Pharmacometrics, 8, 757 (1974)
31) F. Ariyuki et al.: The Clinical Report, 8, 3401 (1974)

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